5. TJEDAN

Fourierov red

Zadatak 1.) FS: $x[n] = \cos\left(\frac{\pi n}{3}\right) + \sin\left(\frac{\pi n}{4}\right)$

$$X_k = \frac{1}{N}\sum_{n=0}^{N-1} x[n]\cdot e^{-2\pi j\frac{kn}{N}}$$

$$\left(\frac{\pi}{3}, \frac{\pi}{4}\right) \rightarrow \frac{\pi}{12} = \Omega_0$$

$$\Omega_0 = \frac{2\pi}{N} = \qquad N = \frac{2\pi}{\pi}\cdot 12 \qquad \boxed{N_0 = 24}$$

$$x[n] = \frac{e^{j\frac{\pi n}{3}} + e^{-j\frac{\pi n}{3}}}{2} + \frac{e^{j\frac{\pi n}{4}} - e^{-j\frac{\pi n}{4}}}{2j}$$

$$x[n] = \frac{1}{2}e^{j\frac{\pi n}{3}} + \frac{1}{2}e^{-j\frac{\pi n}{3}} + \frac{1}{2}e^{-j\frac{\pi}{2}}\cdot e^{j\frac{\pi n}{4}} - \frac{1}{2}e^{-j\frac{\pi}{2}}e^{-j\frac{\pi n}{4}}$$

$$x[n] = \frac{1}{2}e^{j4\Omega_0 n} + \frac{1}{2}e^{\underbrace{-j4}_{-4}\Omega_0 n} + \frac{1}{2}e^{-j\frac{\pi}{2}}\cdot e^{j3\Omega_0 n} + \frac{1}{2}e^{j\frac{\pi}{2}}\cdot e^{\underbrace{-j3}_{-3}\Omega_0 n}$$

$$\underset{(-4)}{1} - \underset{(-3)}{2} - \underset{(-2)}{3} \cdots \underset{(-1)}{20} - 21 - 22 - 23 - 24 \longrightarrow \frac{1}{2}e^{-j\cdot 20\Omega_0 n}$$

$$\longrightarrow \frac{1}{2}e^{j\frac{\pi}{2}}\cdot e^{j21\Omega_0 n}$$

$$\Rightarrow x[n] = \frac{1}{2}e^{j4\Omega_0 n} + \frac{1}{2}e^{j20\Omega_0 n} + \frac{1}{2}e^{-j\frac{\pi}{2}}\cdot e^{j3\Omega_0 n} + \frac{1}{2}e^{j\frac{\pi}{2}}\cdot e^{j21\Omega_0 n}$$

$\longrightarrow X_k$ je onaj broj uz $e^{-jkn\underbrace{\frac{2\pi}{N}}_{\Omega_0}} = e^{-jkn\Omega_0}$

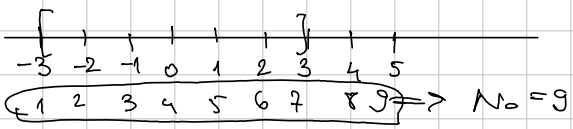
$$\boxed{X_4 = \frac{1}{2} \qquad X_{20} = \frac{1}{2} \qquad X_3 = \frac{1}{2}e^{-j\frac{\pi}{2}} \qquad X_{21} = \frac{1}{2}e^{j\frac{\pi}{2}}}$$

dobiveni spektar je diskr. i periodičan

Snaga signala: $P = \frac{1}{N}\sum_{n=0}^{N-1}|x[n]|^2 = \sum_{k=0}^{N-1}|X_k|^2 = \frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4} = \boxed{1}$

Zadatak 2.) Izračunajte vremenski diskretan Fourierov red periodičnog diskretnog signala, čija je jedna perioda definirana na sljedeći način:

$$x[n] = \begin{cases} n, & |n| \le 3 \\ 0, & n \in \{4, 5\} \end{cases}$$



$$X_k = \frac{1}{9}\sum_{n=0}^{8} x[n]\, e^{-2\pi j k n \frac{1}{9}}$$

$$k=1 \longrightarrow X_k = \frac{1}{9}\left(\sum_{n=-3}^{5} x[n] e^{-jn\cdot\frac{2\pi}{9}k}\right) = \frac{1}{9}\Big[-3\cdot e^{+j3\cdot\frac{2\pi}{9}k} - 2\cdot e^{+j2\cdot\frac{2\pi}{9}k} - e^{+j\frac{2\pi}{9}k} + 0$$

$$+ e^{-j\frac{2\pi}{9}k} + 2e^{-j2\cdot\frac{2\pi}{9}k} + 3e^{-j3\frac{2\pi}{9}k}\Big]$$

$$X_k = \frac{1}{9}\left[-3e^{j\frac{6\pi}{9}k} - 2e^{j\frac{4\pi}{9}k} - e^{j\frac{2\pi}{9}k} + e^{-j\frac{2\pi}{9}k} + 2e^{-j\frac{4\pi}{9}k} + 3e^{-j\frac{6\pi}{9}k}\right]$$

$$X_k = \frac{1}{9}\left[-3\left(e^{j\frac{6\pi}{9}k} - e^{-j\frac{6\pi}{9}k}\right) - 2\left(e^{j\frac{4\pi}{9}k} - e^{-j\frac{4\pi}{9}k}\right) - \left(e^{j\frac{2\pi}{9}k} - e^{-j\frac{2\pi}{9}k}\right)\right]$$

$$X_k = \frac{1}{9}\left[-3\cdot(2j)\sin\left(\frac{6\pi}{9}k\right) - 2\cdot 2j\cdot\sin\left(\frac{4\pi}{9}k\right) - 2j\cdot\sin\left(\frac{2\pi}{9}k\right)\right]$$

$$\boxed{X_k = \frac{-2j}{9}\left(\sin\left(\frac{2\pi}{9}k\right) + 2\sin\left(\frac{4\pi}{9}k\right) + 3\sin\left(\frac{6\pi}{9}k\right)\right)}$$

Snaga signala: $P = \sum_{k=0}^{N-1} |X_k|^2 = \frac{1}{N}\sum_{n=0}^{N-1} |x[n]|^2 =$

$$P = \frac{1}{9}\left((-3)^2 + (-2)^2 + (-1)^2 + 0 + (1)^2 + 2^2 + 3^2\right) = \boxed{\frac{28}{9}}$$

Zadatak 3.) $x(t) = 2\cos(200\pi t) + 3\cos(500\pi t)$

ako se dani signal očita sa $F_s = 1kHz$, nađite koef FS dobivenog diskr signala.

$x(t)$ = periodičan $\rightarrow$ DTFS

$\omega_0 = \frac{2\pi}{T_0}$

$F_s = 1kHz$ $\quad 1kHz > 2f_0$

$\rightarrow T_s = 10^{-3}s$

$$x(nT_s) = 2\cos(200\pi \cdot nT_s) + 3\cos(500\pi n T_s) = 2\cos(200 \cdot \pi \cdot 10^{-3} n) + 3\cos\left(\frac{500}{1000}\pi n\right)$$

$$x[n] = 2\cos\left(\frac{\pi}{5}n\right) + 3\cos\left(\frac{\pi}{2}n\right)$$

DTFS

$$\left(\frac{\pi}{5}, \frac{\pi}{2}\right) = \frac{\pi}{10} \rightarrow \Omega_0 = \frac{2\pi}{N}$$

$$N_0 = \frac{2\pi}{\frac{\pi}{10}} = 20$$

$$x_k = \frac{1}{N}\sum_{n=0}^{N-1} x[n] \cdot e^{-2\pi j \frac{kn}{N}}$$

$$x_k = \frac{1}{20}\sum_{n=0}^{19} x[n] e^{-jkn\frac{2\pi}{20}}$$

$$x[n] \; e^{j\frac{\pi}{5}n} + e^{-j\frac{\pi}{5}n} + \frac{3}{2}\left(e^{j\frac{\pi}{2}n} + e^{-j\frac{\pi}{2}n}\right)$$

$$x[n] = e^{j2\Omega_0 n} + e^{-j2\Omega_0 n} + \frac{3}{2}\left(e^{j5\Omega_0 n} + e^{-j5\Omega_0 n}\right)$$

$-2 = 18 \qquad -5 = 15$

$$x[n] = e^{j2\Omega_0 n} + e^{j18\Omega_0 n} + \frac{3}{2}e^{j5\Omega_0 n} + \frac{3}{2} \cdot e^{j15\Omega_0 n}$$

$$\boxed{x_2 = 1 \quad x_{18} = 1 \quad x_5 = \frac{3}{2} \quad x_{15} = \frac{3}{2}}$$

Snaga: $P = \sum_{k=0}^{N-1} |x_k|^2 = 1 + 1 + \frac{9}{4} + \frac{9}{4} = \frac{13}{2}$

Zadatak 4.) DTFT od $x[n] = \begin{cases} n, & |n| \leq 3 \\ 0, & \text{inače} \end{cases}$

$$X[e^{j\omega}] = \sum_{n=-\infty}^{\infty} x[n] \cdot e^{-j\omega n} = \sum_{n=-3}^{3} x[n] \cdot e^{-j\omega n} = -3 \cdot e^{-j\omega(-3)} + (-2) \cdot e^{-j\omega(-2)}$$

$$+ (-1) e^{-j\omega(-1)} + 0 + e^{-j\omega} + 2e^{-j\omega 2} + 3e^{-j\omega 3}$$

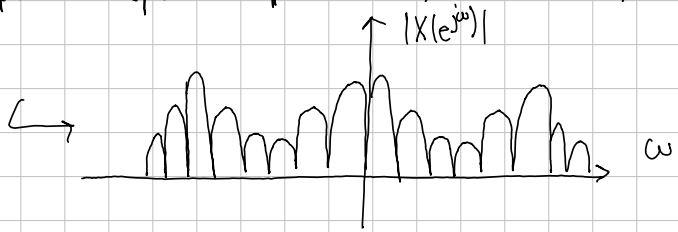
$$X[e^{j\omega}] = -3\left[e^{3j\omega} - e^{-3j\omega}\right] - 2\left(e^{2j\omega} - e^{-2j\omega}\right) - \left(e^{j\omega} - e^{-j\omega}\right)$$

$$X[e^{j\omega}] = 2j\left(-3 \cdot \sin(3\omega) - 2\sin(2\omega) - \sin(\omega)\right)$$

$$X[e^{j\Omega}] = -2j\left(3\sin(3\Omega) + 2\sin(2\Omega) + \sin(\Omega)\right)$$

Totalna energija: $\sum_{n=-\infty}^{\infty} |x[n]|^2 = \boxed{28}$

Amplitudni spektar: $\left|3\sin(3\Omega) + 2\sin(2\Omega) + \sin(\Omega)\right|$



Zadatak 5.) Zadan je spektar $X(e^{j\Omega})$. Nađite signal $x[n]$, te odredite energiju tog signala.

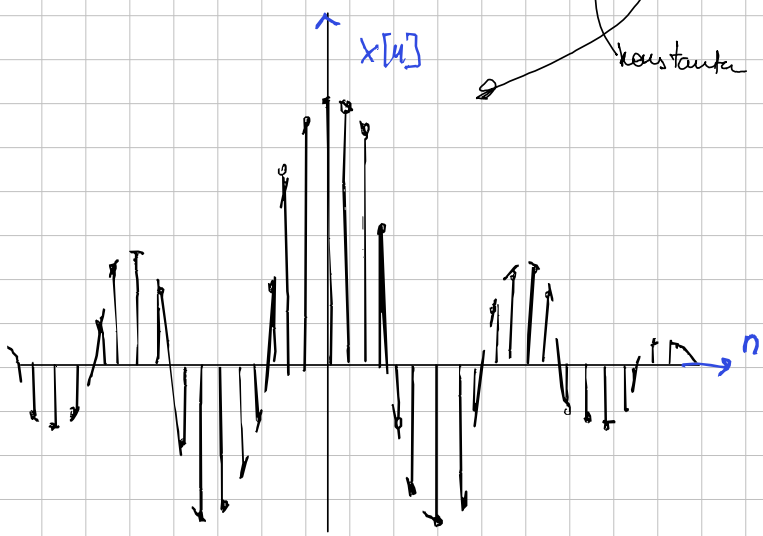
$$X(e^{j\omega}) = \begin{cases} 1, & |\Omega| < \omega \\ 0, & \omega < |\Omega| < \pi \end{cases}$$

$$x[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{j\Omega})\, e^{j\Omega n}\, d\Omega = \frac{1}{2\pi}\int_{-\omega}^{\omega} 1 \cdot e^{j\Omega n}\, d\Omega$$

$$x[n] = \frac{1}{2\pi}\cdot\frac{1}{j\Omega}\, e^{j\Omega n}\Big|_{-\omega}^{\omega} = \frac{1}{2\pi j\Omega}\cdot\left(e^{j\omega n} - e^{-j\omega n}\right) = \frac{1}{\cancel{2}\pi \cancel{j} n}\cdot \cancel{2}\cancel{j}\cdot\sin(\omega n)$$

$$\boxed{x[n] = \frac{\sin \omega n}{\pi n} = \frac{\omega}{\pi}\cdot\frac{\sin \omega n}{\omega n} = \frac{\omega}{\pi}\,\mathrm{sinc}(\omega n)}$$

konstanta



Energija $E = \sum_{n=-\infty}^{\infty}\left(\frac{\omega}{\pi}\cdot \mathrm{sinc}(\omega n)\right)^2 = \left(\frac{\omega}{\pi}\right)^2 \cdot \sum_{n=-\infty}^{\infty} \frac{\sin(\omega n)}{\omega n} = \frac{\omega}{\pi}$ (1)